



Kolmogorov–Arnold Network-Enhanced Temporal Models on Cloud–Edge System for Real-Time Apple Spoilage Monitoring

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Received: 17 November 2025 / Accepted: 18 January 2026
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Abstract

Postharvest losses of apples during storage and transportation pose a significant challenge to food security and economic benefits. This research proposed a framework for apple spoilage detection and prediction that combines multi-gas sensing, deep learning, and a cloud–edge device. Relevant environmental parameters were continuously collected from the storage environment and transmitted to a cloud platform for unified management and model training. Enhanced models LSTM-KAN, BiLSTM-KAN, and TCN-KAN were constructed by integrating the Kolmogorov–Arnold network (KAN) module into the neural network architecture. These models demonstrated higher prediction accuracy and robustness in apple spoilage prediction tasks and significantly improved training convergence speed. Based on multi-sensor array data capturing key physicochemical parameters, the trained models predicted the composite spoilage index (CSI), which quantitatively reflected the degree of apple spoilage, with the TCN-KAN model showing the best performance ($R_p^2 = 0.968$). Furthermore, a cloud–edge device collaboration framework was constructed to support centralized model updates, remote synchronization, and real-time inference. To ensure efficient deployment on resource-constrained devices, a structured pruning technique was developed to reduce model size and improve inference speed while minimizing the loss of prediction accuracy. The pruned TCN-KAN model achieved a 42.68% speed improvement. The proposed method provides an accurate, scalable, and cost-effective approach for real-time monitoring and prediction of spoilage in stored apples, highlighting the potential of deep learning and edge intelligence in smart agriculture applications.

Keywords Apple spoilage prediction · Deep learning · Kolmogorov–Arnold network · Edge computing · Artificial Intelligence of Things

Highlights

- Developed KAN-enhanced models for intelligent apple spoilage prediction.
- KAN-enhanced models accelerated convergence and improved prediction accuracy.
- Proposed an AIoT-based cloud–edge system for real-time spoilage monitoring.
- Applied structured pruning for edge deployment with minimal accuracy loss.
- Established the pruned TCN-KAN as a practical solution for intelligent storage.

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Introduction

Apples are one of the most important horticultural products in global agriculture. However, apples are particularly vulnerable to postharvest losses during storage and transportation (Guo et al., 2023; Osinenko et al., 2021). The scale of these losses is huge, causing obvious economic losses and environmental impacts. Therefore, developing an efficient, intelligent, and responsive apple storage system is of great practical significance for promoting food safety, reducing waste, and optimizing cold chain logistics (Guo et al., 2020; Luo et al., 2026; Praeger et al., 2020; Salamai et al., 2021).

Postharvest quality prediction has advanced from experience-based models to intelligent, data-driven methods. Early analytical methods, including rule-based systems and statistical models, provided an initial framework for quantifying environmental changes. Machine learning methods (Yassen

et al., 2024), such as principal component analysis (PCA), partial least squares regression (PLSR), and support vector machine (SVM), have further improved spoilage prediction by leveraging multivariate data and reducing manual intervention (Fan et al., 2023; Qiao et al., 2025). Recent progress in deep learning (Hu et al., 2025) has enabled more effective modeling of nonlinear time-series relationships in multi-sensor datasets (Khafaga et al., 2022a; Mohd Ali & Hashim, 2024; Sun et al., 2026). Recurrent architectures such as long short-term memory (LSTM) networks (Nunekpeku et al., 2025; Siami-Namini et al., 2019) and convolutional frameworks like temporal convolutional networks (TCN) have been widely applied for postharvest monitoring, achieving superior performance in representing long-term dependencies and local temporal structures (Benidis et al., 2023; Wang et al., 2024; Yang et al., 2025). The recently proposed Kolmogorov–Arnold network (KAN) offers a new perspective for enhancing deep learning models by allocating learning capacity to trainable nonlinear functions along network edges. This structure improves functional approximation, accelerates convergence, and enhances robustness against noise and overfitting (Zou et al., 2025), making it well suited for complex agricultural prediction tasks. Kolmogorov–Arnold network (KAN) is introduced into the temporal neural network model, which is expected to improve the performance of the network model (Liu et al., 2024; Zhu et al., 2025). In KAN, traditional fully connected layers with fixed linear weights are replaced by spline parameterized trainable univariate functions, allowing each connection to capture complex nonlinear relationships. This design enhances the function approximation capability, speeds up convergence, and improves robustness (Li et al., 2025).

Internet of Things (IoT) (Miao et al., 2024), edge computing (Amogi et al., 2024), and artificial intelligence (AI) (Akkas Ali et al., 2024) provide distributed sensing systems that allow a new approach for real-time monitoring and intelligent control of apple storage. The convergence of Internet of Things (IoT) and artificial intelligence (AI) technologies, collectively known as Artificial Intelligence of Things (AIoT) (Jiang et al., 2025), has greatly advanced the digital and intelligent control of fruit storage environments (Botelho et al., 2025). However, deep learning frameworks typically require high computational resources, limiting their real-time use on embedded or low-power devices (Kang et al., 2025). Cloud-centered systems also face latency and communication bottlenecks that reduce responsiveness. Thus, enhancing nonlinear learning capacity, improving computational efficiency, and enabling lightweight deployment on edge devices are crucial for achieving accurate and real-time spoilage prediction in intelligent cold chain systems (Hu et al., 2021). Recent studies have increasingly explored distributed computing frameworks to address the latency, scalability, and resource limitations in intelligent agricultural

systems. Conventional cloud-only architectures often suffer from high communication overhead and limited real-time responsiveness, while device-level computation alone cannot handle the complexity of modern deep learning models (Ren et al., 2019). To overcome these challenges, recent research has adopted hierarchical cloud–edge–device paradigms that balance centralized model management with decentralized inference (Sharma et al., 2025; Tu et al., 2025; Zhang et al., 2024).

This study aims to develop an intelligent monitoring and warning system for apple storage by fusing multi-sensor environmental data with enhanced deep learning models within an AIoT edge computing framework. The main contributions are as follows:

- (1) To enhance model performance, a Kolmogorov–Arnold network module was introduced into the standard LSTM, bidirectional LSTM, and topological recurrent neural network architectures, replacing the traditional fully connected layers with a nonlinear transformation of the inter-layer connections that can be learned. The improvement enhances the model’s ability to fit complex functions, boosts training efficiency, and improves prediction accuracy.
- (2) To support efficient system deployment, a cloud–edge–device collaborative framework was established to support centralized model training, lightweight deployment, and real-time updates.
- (3) To enable edge-oriented optimization, structured pruning techniques were applied to deep learning models to reduce model size and accelerate inference without sacrificing prediction accuracy, thus supporting efficient deployment on low-cost, resource-constrained edge devices.

Materials and Methods

Preparation of Apple Samples

A total of 2000 fresh apples, uniform in size, ripeness, and color, were sourced from a local market near Jiangsu University (Zhenjiang, China) to ensure sample consistency. *Botrytis cinerea*, a primary postharvest pathogen, was employed to simulate spoilage conditions. The strain, obtained from the China Center of Industrial Culture Collection (CICC), was revived and cultured on potato dextrose agar (PDA) at 28 °C for 7 days. Spore suspensions (10^6 cfu/mL) were prepared by harvesting third-generation spores into sterile water. Among the 2000 apples, 1500 were surface-sterilized and inoculated by administering 5 µL of the spore suspension into each of three equatorial punctures on the fruit. The remaining 500 apples were injected with sterile physiological saline and

served as the blank control group. All samples were subsequently incubated at 25 °C and 60% relative humidity.

Apple Storage Sensor Array System

A custom-developed sensor array system was designed to monitor the environmental conditions of stored apple samples. It was designed for real-time monitoring and spoilage detection during apple storage and transportation; the device integrated multiple gas sensors and temperature and humidity sensors to measure key parameters such as volatile organic compounds (VOCs), carbon dioxide (CO₂), oxygen (O₂), temperature, and relative humidity—parameters that have been shown to have a critical impact on apple preservation (Waghmode et al., 2021; Hasan et al., 2024; Zheng & Zhang, 2022). The system demonstrated reliable environmental sensing performance and effectively captured characteristic gas signals associated with apple spoilage. These data streams provide fundamental support for predictive modeling and early warning systems. The selection of sensor modules considered their selectivity, sensitivity, and compatibility with the embedded hardware. The key specifications, including detection range, sensitivity, accuracy, and sensing principle, are summarized in Supplementary Table S1 to ensure experimental clarity and reproducibility. The sensor array was connected to an STM32-based microcontroller for data acquisition and preprocessing. Analog outputs from electrochemical and NDIR sensors were conditioned through filtering and amplification circuits before conversion by the STM32's 12-bit ADC. The temperature and humidity sensor communicated via an I²C interface, and all processed data were transmitted to the edge gateway through a UART serial link for subsequent cloud synchronization. This configuration ensured stable signal acquisition and reliable data transmission across the cloud–edge–device system.

Data Acquisition

A total of 2000 apple samples were used in this study, of which 1800 were inoculated samples and 200 were saline-injected blank control groups. A total of 150 apples were selected from 1800 apples using systematic random sampling, forming one group. Each group was continuously monitored for 10 min at a sampling rate of 1 Hz to record environmental information. The monitored parameters included temperature, humidity, carbon dioxide concentration, oxygen concentration, and volatile organic compounds (VOCs). In total, 300 groups of samples were collected. The measurement period was 10 days and was designed to capture progressive changes in apple spoilage under controlled storage conditions. The resulting environmental data portrayed the different stages of spoilage as microorganisms continued to multiply and the degree of physiological deterioration of apples progressively increased

with longer storage time. Meanwhile, 150 apples were randomly selected from 200 control apples to form a group, and a total of ten groups were measured under the same conditions to establish baseline values for gas concentration and surface color characteristics. Multiple replicate measurements were performed for each group of samples to ensure data robustness and reproducibility.

Considering the inherent biological diversity of agricultural products, each apple sample represented an independent biological individual with unique physiological and biochemical characteristics. Natural variations in soluble solids, firmness, peel structure, and microbial colonization introduced substantial intra-class variability across samples. Continuous monitoring of a large number of apples over time enabled the dataset to capture both biological and temporal diversity, reflecting realistic variability encountered in storage and transportation environments.

Construction of Composite Spoilage Index

To quantify the degree of spoilage for the purpose of spoilage modeling, a composite spoilage index (CSI) was constructed by combining the spoilage area and colorimetric attributes. The observed area of spoilage (S) and color difference (ΔE) were first normalized to the maximum value in all samples, respectively:

$$S_{\text{norm}} = \frac{S}{S_{\text{max}}} \quad (1)$$

$$\Delta E_{\text{norm}} = \frac{\Delta E}{\Delta E_{\text{max}}} \quad (2)$$

Multivariate linear regression was performed using the normalized parameters as predictors and sensory scores (on a 1–5 scale) as the dependent variable. Predictor variables were z -transformed to eliminate scale discrepancies. The standardized regression coefficients obtained from this analysis were used to calculate the weight coefficients:

$$\omega_S = \frac{|\beta_S|}{|\beta_S| + |\beta_{\Delta E}|} \quad (3)$$

$$\omega_{\Delta E} = \frac{|\beta_{\Delta E}|}{|\beta_S| + |\beta_{\Delta E}|} \quad (4)$$

Multicollinearity was evaluated using the variance inflation factor (VIF), ensuring $\text{VIF} < 5$ for all predictors. The final CSI was computed as a weighted sum:

$$\text{CDI} = \omega_S \cdot S_{\text{norm}} + \omega_{\Delta E} \cdot \Delta E_{\text{norm}} \quad (5)$$

This index integrates both morphological and chromatic indicators to quantify apple spoilage severity.

AIoT-Enabled Edge Computing Architecture for Apple Storage Monitoring

To overcome the limitations of high latency, bandwidth consumption, and sluggish response that exist in traditional cloud computing (Vo et al., 2022), this paper proposes an AIoT-based edge computing architecture for smart apple storage monitoring. The framework integrates sensing, edge-level intelligence, and cloud-based optimization to achieve real-time environment monitoring, fast decision-making, and continuous model optimization. Sensor array devices deployed in the apple storage environment are responsible for collecting key parameters such as temperature, humidity, and gas concentration. Warehouse environment data is transmitted to the edge device for local preprocessing and analysis, and the edge performs real-time inference of lightweight predictive models, thus reducing the amount of cloud data transmission, saving bandwidth resources, and improving

response efficiency (Ahmed et al., 2025; Alahe et al., 2025; Eccles et al., 2024).

The AIoT-based apple storage and logistics edge computing framework (Fig. 1) realizes data access, storage, and processing through layered modules. The unified controller manages IoT devices and system variables, and the state interface, method interface, and ModBus driver realize the communication function of the devices and the data interaction between local devices and cloud services. The cloud-edge interface realizes two-way communication: sensor data is continuously uploaded to the cloud, and the lightweight processed model is sent down to the edge. The architecture supports the local operation of models, effectively reducing latency and improving fault tolerance.

The core edge computing architecture of the collaboration framework (Fig. 2) adopts a three-tier structure of cloud-edge-device. The cloud layer is deployed in the Kubernetes cluster within the KubeEdge framework, which is responsible for aggregating multi-node data and retraining the prediction model based on the updated dataset, and then sending the updated model to the edge after a lightweight

Fig. 1 Overall architecture of AIoT-enabled edge computing system for apple storage and logistics

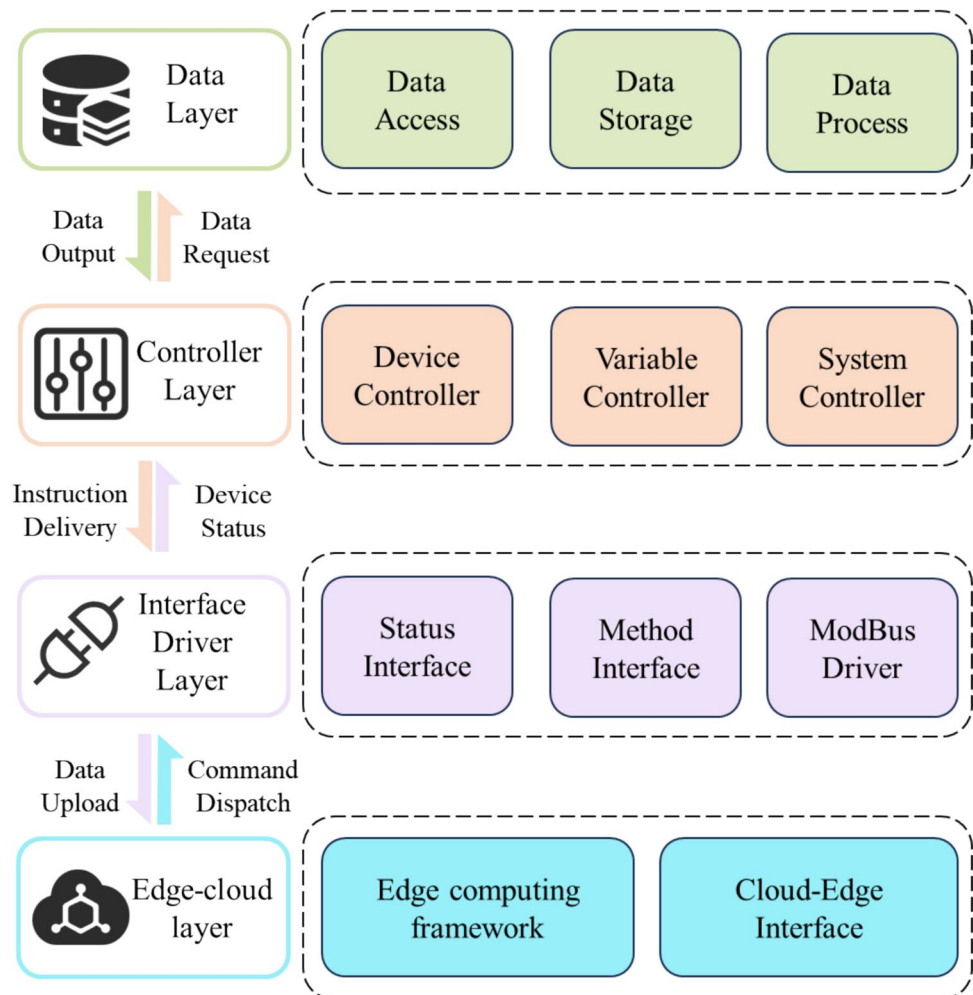
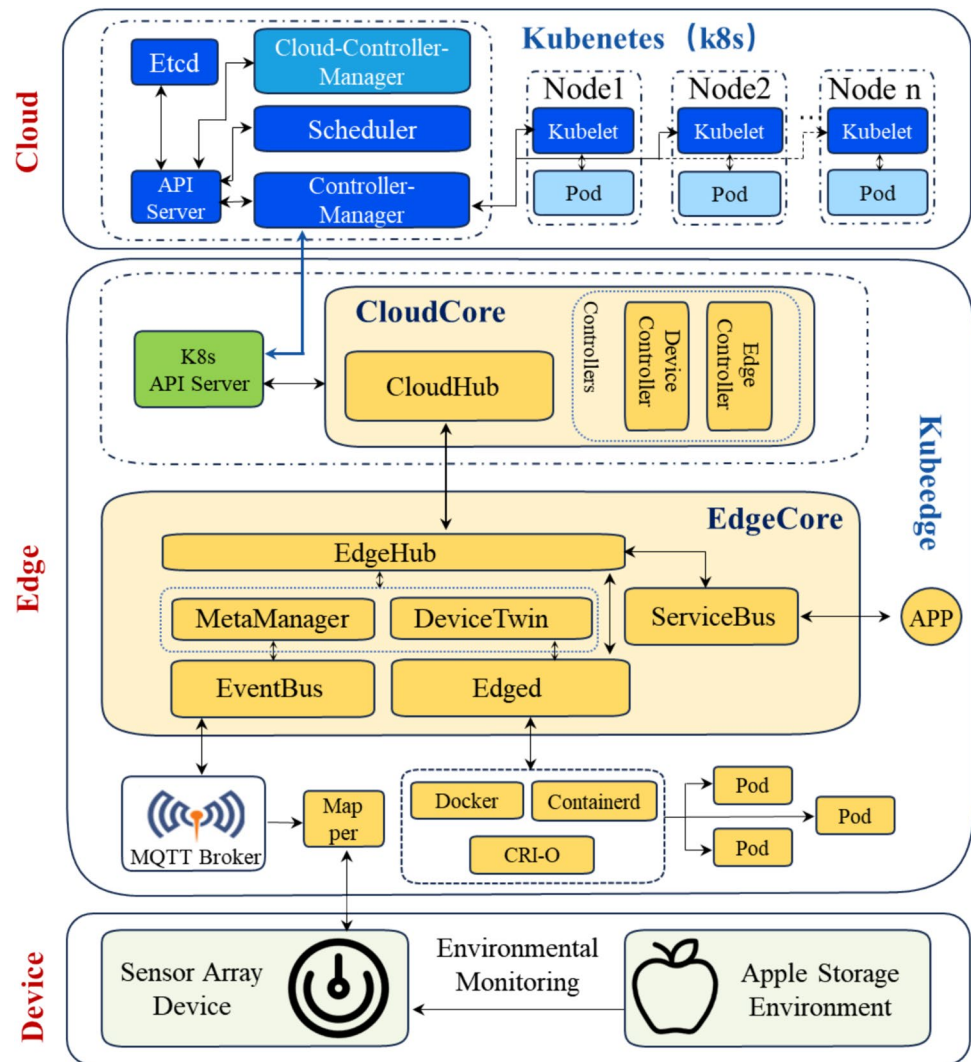


Fig. 2 Edge computing core architecture within the cloud–edge–device collaborative system



operation. The edge layer handles cloud communication through CloudCore and EdgeCore components to realize efficient and stable interaction between the cloud and the edge. It realizes stable execution of computational tasks such as anomaly detection, early warning, and real-time inference at the edge measurement. The device layer, composed of multi-sensor array devices, continuously collects environmental parameters to form a closed-loop system of perception-local decision-making-cloud optimization.

The collaborative framework achieves dynamic adaptability by fusing edge responsiveness with cloud-driven robustness: real-time inference at the edge ensures real-time monitoring and early warning of the apple storage environment, while periodic cloud retraining supports long-term optimization. The architecture provides a scalable, efficient, and reliable solution for smart apple storage management, effectively addressing core challenges such as latency, bandwidth, and system adaptability in distributed agricultural environments.

Deep Learning Models and KAN Integration

To realize an accurate prediction of apple spoilage from sensor array information, a hybrid deep learning framework was developed. The architecture of the proposed model is depicted in Fig. 3. Sequential data collected from storage bins were processed through temporal models (including LSTM, bidirectional LSTM, and temporal convolutional neural networks) to capture dynamic correlations and extract potential representations of spoilage-related patterns (Khafaga et al., 2022b). These extracted features were then fed into either a conventional fully connected layer or a KAN module to generate the final predicted output.

The KAN module is a neural network architecture inspired by the Kolmogorov–Arnold representation theorem, which proves that any multivariate continuous function can be represented as a finite combination of univariate functions and summation operations. Unlike multilayer perceptrons (MLPs), which utilize fixed activation functions at nodes and linear

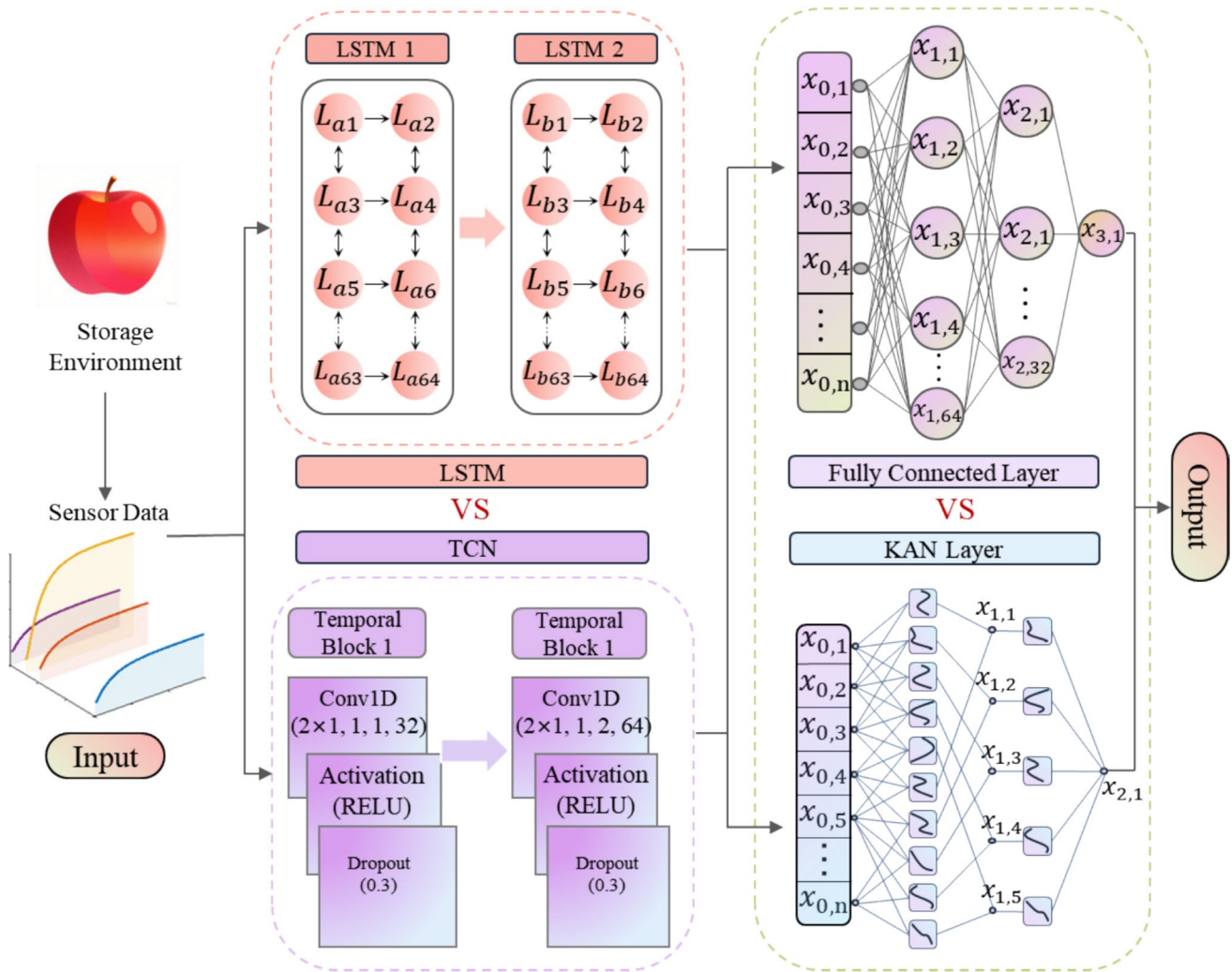


Fig. 3 Architecture of LSTM and TCN models integrated with fully connected and KAN layers

weights along edges, KANs parameterize each edge with a learnable univariate function, commonly implemented as a B-spline. Nonlinear transformations are therefore moved from nodes to edges, enhancing both functional flexibility and interpretability.

Formally, the Kolmogorov–Arnold representation for a function $f : [0, 1]^n \rightarrow \mathbb{R}$ is given by the following:

$$f(x) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \varphi_{q,p}(x_p) \right) \quad (6)$$

where $\varphi_{q,p}$ and Φ_q are univariate continuous functions. This motivates the KAN layer, defined as a matrix of such functions with neuron activations computed as pure summations:

$$x_{l+1,j} = \sum_{i=1}^{n_l} \varphi_{l,j,i}(x_{l,i}) \quad (7)$$

Each function is parameterized as follows:

$$\varphi(x) = \omega_b \cdot \text{silu}(x) + \omega_s \cdot \sum c_k B_k(x) \quad (8)$$

where $\text{silu}(x)$ is a residual basis function, $B_k(x)$ are B-spline basis functions, and c_k are trainable coefficients. The parameters ω_b and ω_s modulate the contribution of each component.

The full forward pass of a KAN with LLL layers is given by the recursive composition:

$$\text{KAN}(x) = (\Phi_{-(L-1)} \circ \Phi_{-(L-2)} \circ \dots \circ \Phi_0)(x) \quad (9)$$

where each Φ_l is a matrix of univariate functions $\varphi_{l,j,i}$, and the activation at layer $l+1$ is obtained by the following:

$$X_{l+1} = \Phi_L(X_l) = \left[\sum_{i=1}^{n_l} \varphi_{l,1,i}(x_{l,i}), \dots, \sum_{i=1}^{n_l} \varphi_{l,n_{l+1},i}(x_{l,i}) \right]^T \quad (10)$$

KANs achieve faster approximation rates than MLPs under smooth compositional assumptions, as formalized by the scaling law:

$$\|f - \hat{f}_G\| C^m \leq CG^{-k-1+m} \quad (11)$$

where G is the number of spline grid intervals, k is the spline order, $m \leq k$ is the smoothness, and C is a constant that depends on the function and its representation. This implies that KANs are able to overcome the curse of dimensionality provided that the smooth combinatoriality assumption is satisfied. KANs remain fully differentiable and can be trained by backpropagation. Their spline functions are initialized with low variance and adapted by gradient descent, while the explicit spline structure of edge activation also endows KANs with interpretability.

In the hybrid architecture, the KAN module replaces the traditional fully connected layer to which the deep learning model is ultimately connected. This combination enables the model to more flexibly and accurately characterize the nonlinear relationship between extracted temporal features and metamorphic correlation metrics. By combining LSTM, bidirectional LSTM, and TCN for sequence feature extraction with spline function-based KAN mapping, complex dependencies in sensor data can be characterized more effectively while maintaining interpretability and stable training behavior.

Structured Model Pruning for Resource-Constrained Edge Deployment

To optimize real-time inference on low-cost, computationally limited embedded edge devices, this study employed a structured model pruning strategy (Cheng et al., 2024; He & Xiao, 2024) to lighten six deep learning architectures: the LSTM, the BiLSTM, the TCN, and its corresponding KAN-enhanced variants. The goal is to reduce the model size and improve the model inference speed without seriously affecting the prediction accuracy.

Unlike unstructured pruning whose removal of individual weights often leads to irregular sparse matrices incompatible with hardware acceleration, structured pruning is used in this study (Fang et al., 2023; Wen et al., 2020). Specifically, all weight parameters of the fully connected and convolutional layers are included in the pruning; in recurrent neural networks (LSTM and BiLSTM), the weight matrices of the input-hidden and hidden-hidden layers are additionally included. For the KAN-augmented model, only the fully connected components are pruned, and the spline function-based KAN edges are retained to maintain its nonlinear transformation capability.

A magnitude-based importance assessment method was used to remove the 30% weights with the lowest magnitude

among all specified layers by global L1-paradigm pruning. Post-pruning removal reparametrizes the wrapper to recover a compact weight matrix, ensuring compatibility with commonly used edge inference libraries. This retraining step is important to maintain learned temporal patterns and feature representations while reducing computational overhead.

Model Training and Evaluation Metrics

All models were trained using the mean square error (MSE) loss function. Model performance was assessed by the coefficient of determination (R_c^2 for calibration, R_p^2 for prediction) and the root mean square error (RMSEC for calibration, RMSEP for prediction). Together, these metrics reflect the fitting effect during training and the prediction accuracy for unknown data, providing a reliable basis for assessing model performance.

To ensure independent and unbiased performance evaluation, a sample-level independent validation approach was adopted. Data from the same apple were never included simultaneously in both the training and test sets, thereby avoiding temporal data leakage and ensuring statistical independence between datasets. This validation design provides a more rigorous evaluation of model generalization.

To mitigate potential overfitting, several regularization and optimization strategies were incorporated during model training. Dropout layers ($p=0.2$) were applied to recurrent and convolutional blocks to reduce neuron co-adaptation, and L2 weight spoilage ($\lambda = 1 \times 10^{-4}$) was added to all trainable parameters.

Results and Discussion

Gas Emission Dynamics During Apple Spoilage

The relationship between apple spoilage process and gas emissions was obtained from the analysis of environmental data collected by the sensor array system during apple storage. The results showed that with the aggravation of apple spoilage, the concentrations of ethylene (C_2H_4), carbon dioxide (CO_2), and volatile organic compounds (VOCs) have continued to rise (Butkeviciute et al., 2022; Xu et al., 2021). In the early stage of spoilage, the color difference (ΔE) value of the affected area also increased significantly, indicating visible changes in the surface.

Ethylene is a key phytohormone involved in fruit ripening, which has a significant impact on the spoilage process of apples. Increased ethylene concentration will promote the degradation of the cell wall, accelerate tissue softening and decomposition, and eventually lead to an increase in the area of spoilage and an aggravation of the degree of spoilage. In addition, ethylene also affects the texture and

appearance of the fruit, which may enhance the chromaticity of the a^* (red–green) and b^* (yellow–blue) channels in the CIELAB color space.

VOCs, commonly released during microbial degradation and tissue breakdown, were present at much higher levels in severely spoiled apples. Their increased concentration was likely correlated with microbial activity and the extent of biochemical spoilage. Furthermore, varied CO_2 levels across different storage conditions were observed to influence apple skin coloration, suggesting that environmental gas composition plays a significant role in altering the visual quality of stored apples.

These findings demonstrated the capability of the sensor array system to effectively capture dynamic gas emission patterns associated with apple spoilage. By monitoring changes in ethylene, CO_2 , and VOC levels in conjunction with visual indicators such as color change, the system provides a foundation for the development of predictive models and early warning mechanisms in intelligent fruit storage applications.

Analysis of the Composite Spoilage Index (CSI) for Apple Spoilage

To quantitatively assess the degree of apple spoilage during storage, a composite spoilage index (CSI) was developed by combining two measurable indicators, spoilage area (S) and color difference (ΔE). The maximum values of these two parameters for all samples were first normalized to eliminate scale differences. To account for potential covariates and ensure interpretability, a z -value transformation was performed on the normalized values, followed by a multiple linear regression using sensory scores (scale 1–5) as the response variable.

The regression analysis yielded standardized coefficients of $\beta_S = 0.768$ and $\beta_{\Delta E} = 0.440$, indicating that the spoilage area had a stronger influence on perceived spoilage compared to chromatic change. Weight coefficients for CSI were calculated based on the absolute values of the standardized coefficients, resulting in a final formulation:

$$\text{CDI} = 0.636 \cdot \frac{S}{S_{\max}} + 0.364 \cdot \frac{\Delta E}{\Delta E_{\max}} \quad (12)$$

Variance inflation factor (VIF) values for both predictors were below 5 (VIF = 4.29), confirming the absence of severe multicollinearity. The derived CSI served as a composite index reflecting both morphological and visual aspects of apple degradation and was used as the prediction target in all subsequent deep learning models.

Comparative Analysis of Model Performance

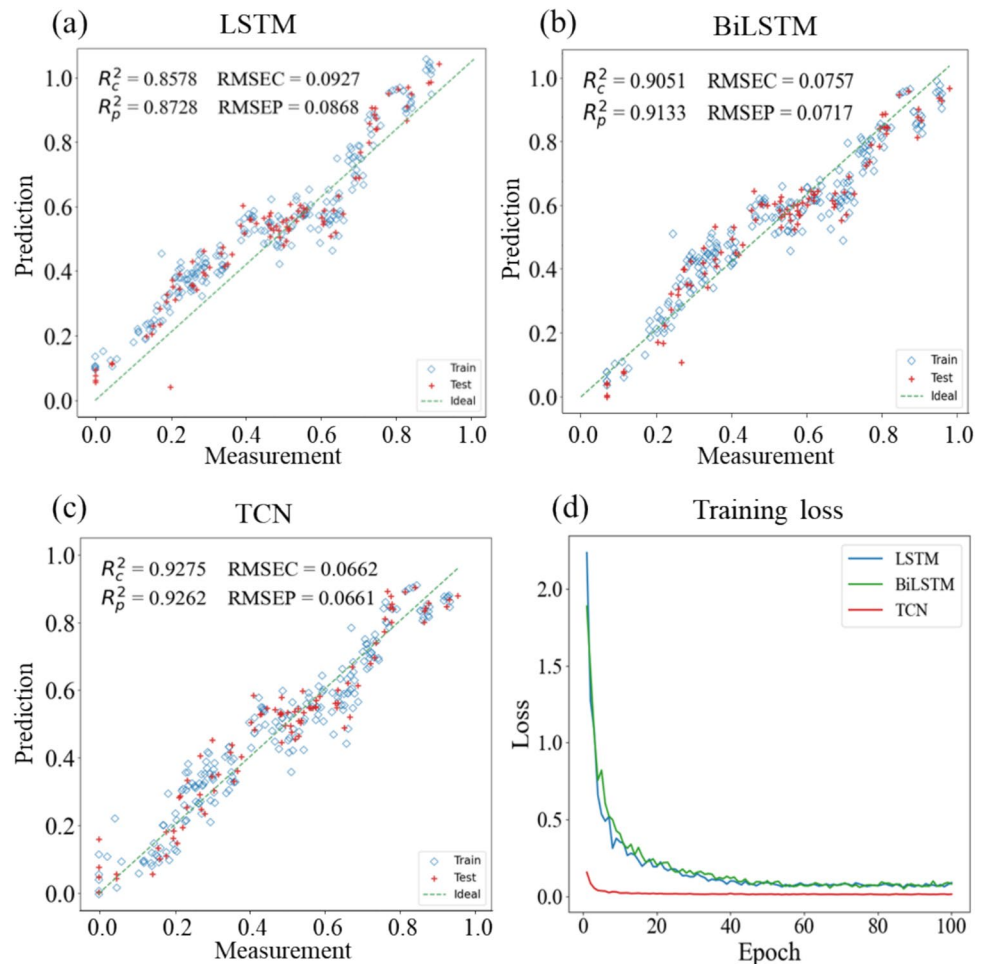
To strictly evaluate the predictive ability of different deep learning architectures to capture the dynamics of apple spoilage, six models were studied, including traditional recurrent and convolutional structures (LSTM, BiLSTM, and TCN) and their KAN-enhanced variants (LSTM-KAN, BiLSTM-KAN, and TCN-KAN). The evaluation was based on multiple indicators, including the coefficients of determination (R_c^2 and R_p^2) and the convergence behavior reflected in the training loss curve. In addition, the scatter distribution of the predicted and measured values was analyzed to assess the deviation, clustering trend, and the degree of coincidence with the regression line, so as to gain a deeper understanding of the predictive advantages and limitations of each model. The results of the traditional model are shown in Fig. 4, while those for the KAN-enhanced models are shown in Fig. 5. Detailed numerical indicators are summarized in Table 1.

Performance of Conventional Models

The prediction ability of LSTM was relatively weak, and the coefficients of determination were $R_c^2 = 0.8578$ and $R_p^2 = 0.8728$, respectively. The RMSEC and RMSEP values (0.0927 and 0.0868, respectively) highlighted their relatively large prediction errors (Table 1). It can be seen from the scatter distribution in Fig. 4a that although the overall trend was close to the ideal regression line, the deviation was more obvious within the range of the medium degree of spoilage (the normalization scale was 0.3–0.6). In this range, the forecast values tended to be more widely dispersed, which reflected the limitations of LSTM in capturing nonlinear changes between the early and late stages of spoilage. The abnormal value of the deviation from the regression line showed that it was difficult to generalize the model to different attenuation patterns. In addition, the training loss trajectory in Fig. 4d showed a slow convergence speed, and the loss value remained above 0.5 in a considerable number of training rounds, which confirmed that LSTM required longer training to extract time dependence. This phenomenon was consistent with the dependence of the gate control mechanism on sequential propagation, and sequential propagation hindered effective characterization learning in data sets with irregular time dynamic characteristics.

By introducing bidirectional time processing, BiLSTM significantly improved the prediction accuracy. The scattering point was more consistent with the regression line, and the systematic deviation in each stage of spoilage was smaller (Fig. 4b). This improvement was reflected in the higher coefficients of determination ($R_c^2 = 0.9051$, $R_p^2 = 0.9133$) and reduced errors (RMSEC = 0.0757, RMSEP = 0.0717). However, a careful examination of the

Fig. 4 Performance metrics of three baseline models. **a** LSTM predictions, **b** BiLSTM predictions, **c** TCN predictions, **d** Comparative training loss of models



scatter distribution showed that there was still residual proliferation at a high level of spoilage (>0.7), indicating that the model underestimated the degree of spoilage in some cases. The training loss curve in Fig. 4d also showed that the convergence speed of the bidirectional LSTM was faster than that of the unidirectional LSTM, and it stabilized below 0.2 after about 30 epochs. These results emphasized that bidirectional processing contributed to more comprehensive contextual learning, but the sensitivity to small fluctuations indicated that it was more sensitive to the noise in the data set.

The performance of TCN was better than that of the two recurrent structures. As shown in Fig. 4c, the scattering points were densely concentrated on the ideal regression line, and the degree of dispersion was extremely low at all degrees of spoilage. Quantitative indicators further confirmed this point: $R_c^2 = 0.9275$ and $R_p^2 = 0.9262$, as well as lower RMSEC (0.0662) and RMSEP (0.0661), all of which indicated the prediction accuracy of the model. Compared with LSTM and BiLSTM, there were few outliers and the deviation was smaller. Importantly, the model showed robustness in predicting extreme spoilage, superior to other

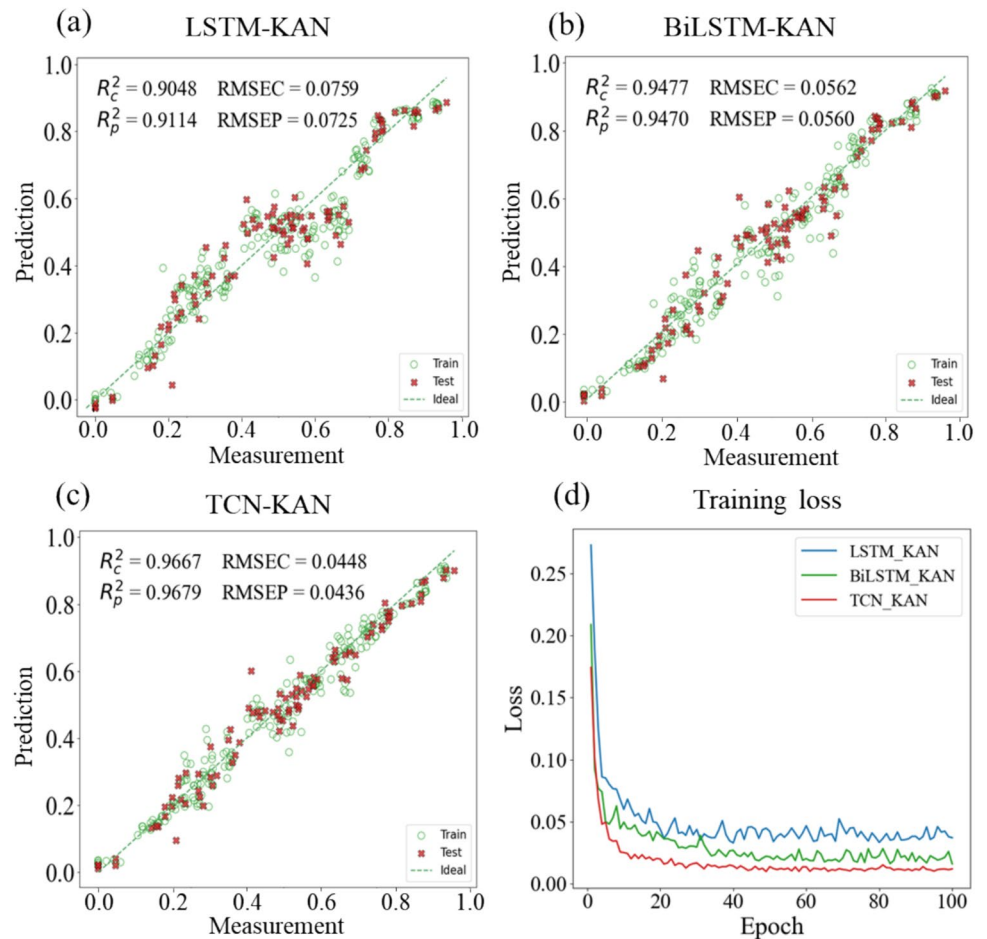
architectures that tended to underestimate or overvalue in similar situations. The training loss trajectory in Fig. 4d confirmed rapid convergence, and the loss decreased sharply in the first 10 epochs and stabilized at a level below the two cycle models. This reflected that convolutional design could effectively capture local and long-range dependencies and avoided the common vanishing gradient problem in recurrent models.

Performance of KAN-Enhanced Models

The integration of the KAN into the traditional model significantly improved the performance of all indicators. By introducing flexible function representation, KAN improved the ability of the model to approximate nonlinear spoilage dynamics, reduced the system deviation and stochastic variance in the prediction, and improved the accuracy of the prediction.

The LSTM-KAN model showed significant improvements compared with the traditional LSTM model. As shown in Fig. 5a, the scattered distribution was more closely concentrated around the regression line,

Fig. 5 Performance metrics of three KAN-enhanced models. **a** LSTM-KAN predictions, **b** BiLSTM-KAN predictions, **c** TCN-KAN predictions, **d** comparative training loss of models



especially in the intermediate spoilage range, where the baseline LSTM model had a large deviation. The coefficients of determination increased ($R_c^2 = 0.9048$, $R_p^2 = 0.9114$), while the errors decreased (RMSEC = 0.0759,

RMSEP = 0.0725). This reduction in dispersion reflected the ability of KAN to enhance the characterization ability of LSTM, making up for its limitations in terms of sequence. The training loss curve in Fig. 5d further showed that it converged faster, and the loss value dropped below 0.1 in early training rounds, which contrasted with the slower decline observed in the traditional LSTM.

BiLSTM-KAN further improved the prediction accuracy. Scattering points were tightly gathered near the regression line, and the degree of dispersion in all stages of spoilage was very small (Fig. 5b). The performance indicators were significantly improved ($R_c^2 = 0.9477$ and $R_p^2 = 0.9470$, while RMSEC (0.0562) and RMSEP (0.0560) were also significantly reduced. Compared with the baseline BiLSTM, the prediction bias at a higher degree of spoilage was greatly reduced. In addition, the training loss curve in Fig. 5d showed rapid convergence and excellent stability, indicating that KAN not only improved the characterization accuracy, but also regularized the optimization process, thus alleviating the overfitting tendency.

TCN-KAN achieved the best overall performance. The scatter distribution in Fig. 5c almost coincided with the ideal regression line, indicating that the model had the tightest

Table 1 Comparative performance of baseline, KAN-enhanced, and pruned models on calibration and prediction sets

Model	Calibration set		Prediction set	
	R_c^2	RMSEC	R_p^2	RMSEP
LSTM	0.8578	0.0927	0.8728	0.0868
LSTM-pruned	0.7974	0.1107	0.8252	0.1018
BiLSTM	0.9051	0.0757	0.9133	0.0717
BiLSTM-pruned	0.9005	0.0776	0.9097	0.0731
LSTM-KAN	0.9048	0.0759	0.9114	0.0725
LSTM-KAN-pruned	0.9394	0.0605	0.9359	0.0616
BiLSTM-KAN	0.9477	0.0562	0.9470	0.0560
BiLSTM-KAN-pruned	0.9275	0.0662	0.9251	0.0666
TCN	0.9275	0.0662	0.9262	0.0661
TCN-pruned	0.8608	0.0918	0.8671	0.0887
TCN-KAN	0.9667	0.0448	0.9679	0.0436
TCN-KAN-pruned	0.9244	0.0676	0.9287	0.0650

clustering and the least abnormal values among all the evaluated models. The coefficients of determination reached the highest level ($R_c^2 = 0.9667$, $R_p^2 = 0.9679$), while RMSEC and RMSEP were reduced to 0.0448 and 0.0436, respectively, indicating excellent prediction accuracy. Unlike other models, the TCN-KAN model showed consistent accuracy in the early and late stages of spoilage, highlighting its good generalization ability. The training loss trajectory in Fig. 5d also confirmed the rapid and stable convergence and obtained the lowest final loss value. These findings emphasized the synergistic advantage of combining TCN's efficient time feature extraction with KAN's nonlinear approximation ability, thus building an architecture that simulated the spoilage process with high accuracy.

Training Convergence Behavior

The training loss curves (Fig. 6) provided supplementary information for the previously discussed accuracy index and scatter distribution. Although the previous analysis emphasized the improvement of accuracy, error reduction, and regression line alignment, the convergence trajectory revealed the underlying optimization dynamics that drove these improvements.

All architectures observed a consistent trend: the introduction of KAN not only accelerated convergence, but also reduced the final loss value. For LSTM (Fig. 6a), the baseline model showed a slow and unstable decline, and the loss value remained above 0.5 in a considerable number of iteration cycles. In contrast, LSTM-KAN quickly reduced the training loss to less than 0.1 in the initial iteration, and then stabilized with very little fluctuation, which reflected the improvement of its characterization reliability.

The BiLSTM (Fig. 6b) was faster than that of the unidirectional LSTM, but it still fluctuated in later iterations. Through KAN integration, BiLSTM-KAN not only significantly reduced the loss value, but also showed a smoother

convergence process, which was consistent with the excellent predictive stability of the previously reported model. This showed that KAN played the role of both a functional enhancer and a regularizer, thus alleviating the over-simplification and reducing the sensitivity to the noise attenuation mode.

For the TCN architecture (Fig. 6c), because the dilated convolution could efficiently capture the hierarchical time characteristics, its convergence speed was already very fast. However, TCN-KAN achieved further reduction in the final plateau period, which was consistent with its higher prediction accuracy and lower error rate, confirming the advantage of combining convolutional time modeling with KAN's nonlinear expression ability (Jagtap et al., 2020).

Comparative and Ablation Analysis of Model Performance

A comprehensive comparative analysis of the six models was made to comprehensively compare the advantages and disadvantages of the models' performance and the model optimization after the introduction of the KAN network. In the traditional architecture, the performance of TCN was better than that of LSTM and BiLSTM, which reflected the advantages of the convolutional structure in effectively capturing short-range and long-range time dependencies. Although LSTM and BiLSTM showed acceptable performance, the scatter distribution was relatively discrete, especially near lower and higher CSI.

After the introduction of KAN, the performance of all three baseline models was improved, confirming its role in enhancing nonlinear mapping and improving functional expression ability. These improvements were particularly significant in LSTM, and KAN effectively enhanced the predictive performance of the model. For BiLSTM, KAN strengthened bidirectional context processing to improve accuracy and stability. However, the most significant improvement was reflected in TCN-KAN. The introduction

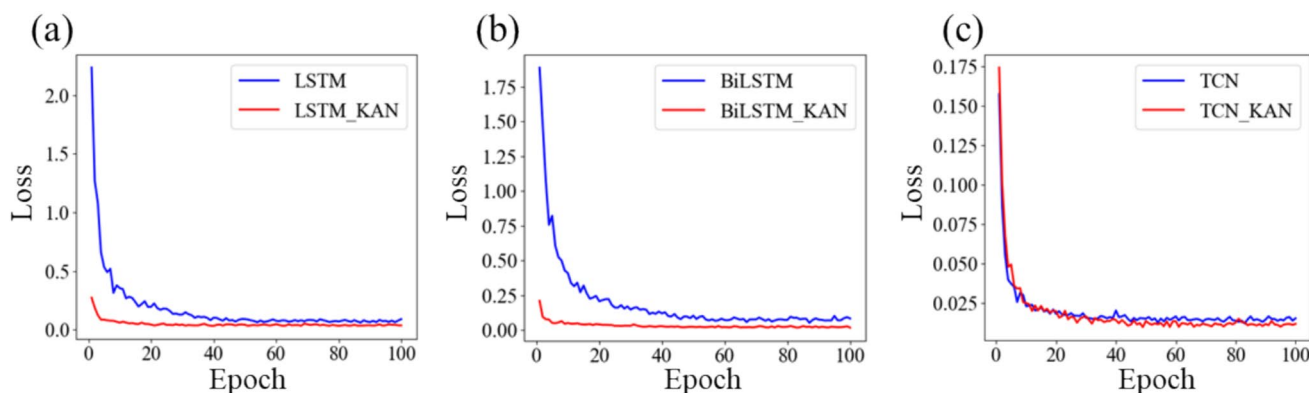


Fig. 6 Training loss curves before and after KAN network integration. **a** LSTM vs. LSTM-KAN, **b** BiLSTM vs. BiLSTM-KAN, **c** TCN vs. TCN-KAN

of KAN amplified the inherent advantages of convolutional architecture, achieving the highest accuracy, minimum error, and the best generalization ability.

The integration of KAN into the traditional architecture not only improved the prediction accuracy, but also significantly improved the training convergence behavior. In the LSTM model, the convergence speed of model training was slow and unstable, and the loss value remained at a high level during the longer training cycle. In contrast, LSTM-KAN achieved rapid and stable convergence, which significantly reduced losses in the early stage of training. Similarly, BiLSTM-KAN also showed smoother and more consistent dynamic characteristics than BiLSTM, indicating that KAN was both a function enhancer and an implicit regularizer, which could effectively alleviate fluctuations in the optimization process. Even for TCN, which was already very efficient, the introduction of KAN further reduced the final loss plateau period and stabilized convergence, which highlighted that KAN's advantages were not only reflected in the characterization ability, but also in the improvement of training efficiency. These improvements in the loss trajectory were consistent with the reduction of prediction errors and closer scatter alignment, highlighting that KAN not only enhanced the expression ability of the model, but also improved its learning stability and robustness.

In addition to the comparative performance evaluation, an ablation analysis was conducted to assess the contribution of the Kolmogorov–Arnold network module. The baseline models, LSTM, BiLSTM, and TCN, along with their KAN-enhanced versions, were trained under identical datasets and hyperparameter settings. As summarized in Table 1, the addition of the KAN layer consistently improved the coefficient of determination R_p^2 and reduced the root mean square error of prediction (RMSEP) across all architectures. On average, R_p^2 increased by about 0.038, while RMSEP decreased by 0.0175. The most significant improvement was observed in the TCN architecture, where R_p^2 increased by 0.0417 and RMSEP decreased by 0.0225. These results confirm that the spline-based representation of the KAN module enhances nonlinear feature extraction and strengthens model generalization by acting as an implicit regularizer.

In summary, KAN brought consistent benefits in all models, and it was most effective when combined with TCN. TCN-KAN achieved the best scatter alignment, minimized the abnormal value, and showed good convergence efficiency, making it the best prediction framework among the six models. These findings emphasized the importance of architectural collaboration: convolutional time modeling provided a robust structural foundation, and KAN introduced the nonlinear representation flexibility required to capture the complex heterogeneous characteristics of the spoilage process. The final prediction model took into account accuracy and generalization ability and had great

practical application potential for real-time monitoring in the cloud–edge collaboration system.

Performance Analysis of Pruned Models

Predictive Performance of Pruned Models

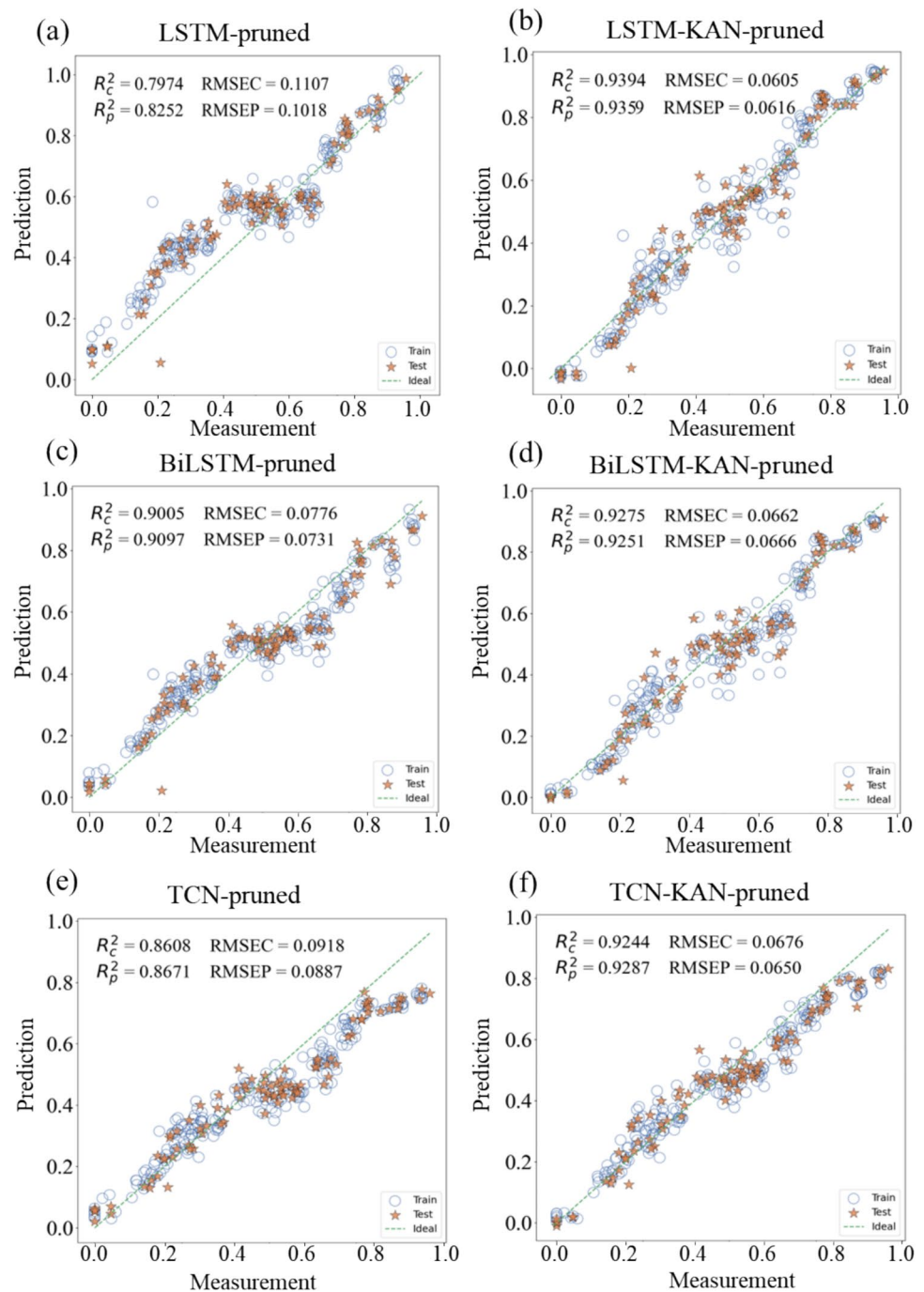
To systematically evaluate the impact of pruning and KAN integration on the architecture of recurrent neural networks and convolutional neural networks, this section compared the performance of LSTM, BiLSTM, and TCN models and their corresponding KAN enhancement models. The analysis considered the prediction accuracy, the robustness of pruning, and the computing efficiency, so as to fully understand how KAN helped balance the accuracy and inference speed of different network structures.

As shown in Fig. 7a, the pruned LSTM model showed weak predictive performance, and there was a significant deviation between the scattering point and the regression line. This deviation was especially obvious at low CSI values, when the predicted value was always low. The test point also showed greater dispersion at high CSI values, reflecting the instability of the model when simulating the late spoilage stage. These deviations were reflected in the performance indicators, $R_p^2 = 0.8252$, RMSEP=0.1018, and there was an obvious gap between calibration accuracy and prediction accuracy. Inference efficiency was significantly improved, and the execution time was reduced from 1.495 to 0.599 s, and the speed increased by 59.93% (Table 2). The results confirmed that although pruning effectively accelerated the calculation, the traditional LSTM architecture lacked robustness to parameter reduction, resulting in a significant decrease in accuracy.

The incorporation of KAN substantially alleviated this problem. As shown in Fig. 7b, in the entire CSI range, the scatter distribution was closely gathered near the regression line, and the variance of the training set and the test set was significantly reduced. The predicted correlation reached $R_p^2 = 0.9359$ with RMSEP=0.0616, indicating that the distortion caused by pruning was effectively compensated. The observed performance improvement came from the spline-based transformation introduced by KAN, which enhanced the nonlinear characterization ability of the LSTM trunk network. Although the reduction in inference time was not large (35.59%), from 1.017 to 0.655 s (Table 2), the comprehensive performance of the model was improved, because it maintained the accuracy of the prediction while maintaining acceptable computational efficiency.

Similarly, the pruned BiLSTM also showed stable prediction ability. As shown in Fig. 7c, the scatter points were densely distributed on the regression line. Although a slight variance was observed at low CSI values, the variability of the test sample was greater than that of the training data

Fig. 7 Predictions of comprehensive spoilage index (CSI) by pruned lightweight models. **a** LSTM-pruned, **b** LSTM-KAN-pruned, **c** BiLSTM-pruned, **d** BiLSTM-KAN-pruned, **e** TCN-pruned, **f** TCN-KAN-pruned



at this time. The overall forecast quality was still satisfactory. $R^2_p = 0.9097$, RMSEP=0.0731, which highlighted that the bidirectional architecture could still integrate past and future dependencies even after pruning. However, the improvement of calculation efficiency was very small, and the inference time was only reduced from 0.510 to 0.459 s, and the increase was only 10.00% (Table 2). This limited acceleration reflected the compact design of BiLSTM, which led to fewer redundant parameters, and the model pruning improved its computing speed less.

The pruned BiLSTM-KAN further reinforced robustness. The degree of coincidence between the scatter points and the regression line was higher, and the residual was symmetrically distributed on the CSI spectrum, which confirmed the improvement of the generalization ability of the model (Fig. 7d). The performance index showed that $R^2_p = 0.9251$ and RMSEP=0.0666, which was significantly improved compared with BiLSTM. The KAN module also offset the impact of pruning and retained the nonlinear feature mapping. The inference time was reduced from 1.441

Table 2 Inference time reduction and speedup ratios before and after structured pruning for all model variants

Model	Before pruning (s)	After pruning (s)	Speedup (%)
LSTM	1.495	0.599	59.93%
LSTM-KAN	1.017	0.655	35.59%
TCN	2.633	1.616	38.63%
TCN-KAN	2.071	1.187	42.68%
BiLSTM	0.510	0.459	10.00%
BiLSTM-KAN	1.441	1.067	25.95%

to 1.067 s (Table 2), and the speed increased by 25.95%. Although the computing efficiency was still lower than that of LSTM-KAN and TCN-KAN, the stability of the prediction results in each stage of spoilage showed that combining KAN with a bidirectional loop architecture was advantageous.

In contrast, the pruned TCN achieved satisfactory performance. Scattering points were generally distributed along the regression line, but there was an obvious deviation at the extreme CSI value, and the predicted value and the measured value were different, especially in the test data set (Fig. 7e), indicating that pruning weakened the ability of the model to extrapolate the boundary situation. However, pruning effectively improved efficiency, with the inference time reduced from 2.633 to 1.616 s, and the speed increased by 38.63% (Table 2). This showed that the convolutional structure of TCN contained considerable redundancy, so it could be pruned effectively with moderate performance loss.

The pruned TCN-KAN model showed the best performance. The scatter distribution was closely arranged along the regression line in the entire CSI spectrum, and the deviation between the training set and the test set was extremely small (Fig. 7f). Various indicators confirmed this stability, $R_p^2 = 0.9287$ and RMSEP = 0.0650. The impact of the outlier value was significantly reduced, highlighting that the smoothing effect of the spline-based KAN module could still retain nonlinear feature extraction after pruning. The inference time was shortened from 2.071 to 1.187 s (Table 2), and the speed increased by 42.68%. Among all post-pruning traditional models and KAN-enhanced models, this was one of the most balanced schemes between accuracy and computational efficiency.

By systematically eliminating redundant parameters and structures, pruning significantly improved the inference speed while reducing the overall scale of the deep learning model. This allowed advanced neural networks to be deployed more efficiently on resource-limited edge devices. The resulting lightweight model could respond to deterioration predictions in real time, reduce hardware and energy requirements, and enhance the scalability of intelligent

warehousing and logistics systems, thus improving its practical application and long-term sustainability.

Comparative Evaluation and Implications of Pruning

The comparative evaluation of the six pruning models revealed three core conclusions. First of all, pruning always improved computing efficiency, but the degree of acceleration depended on the model architecture. Simple recurrent models such as LSTM benefited the most (59.93%), while BiLSTM had the least gain (10.00%) due to its relatively compact structure and fewer redundant parameters. The convolutional model, especially TCN, showed significant acceleration, which reflected the inherent high redundancy in its filter-based design.

The integration of KAN modules consistently reduced the adverse impact of pruning on prediction accuracy. In all cases, the KAN-enhanced model was better than its traditional corresponding model, obtaining higher R_p^2 values and lower RMSEP. The spline-based transformation enriched the nonlinear representation of the spoilage dynamics, thus retaining the ability to extract features that would have been weakened by pruning. The scatter distribution further showed that the KAN-based model maintained a closer match with the regression line at low CSI and high CSI levels, confirming its enhanced stability.

The scatter chart highlighted the different effects of pruning on the CSI spectrum. Ordinary models usually underestimated low CSI values and showed greater dispersion at high values, indicating that pruning disproportionately reduced the sensitivity of models to early subtle spoilage patterns and extreme spoilage processes. On the contrary, the KAN-enhanced variant maintained the prediction accuracy throughout the range, showing the robustness of this deviation.

Overall, the LSTM-KAN architecture after pruning performed the best, achieving the strongest predictive correlation ($R_p^2 = 0.9359$) and the lowest RMSEP (0.0616), while the inference time was reduced by 35.59%. In contrast, the pruned TCN-KAN model achieved the best balance between accuracy and efficiency. It had both stable predictive performance ($R_p^2 = 0.9287$) and the highest computational acceleration (42.68%). Although BiLSTM-based models were also quite accurate, their limited acceleration performance showed that they might not have been suitable for real-time applications with critical efficiency requirements.

To further evaluate the deployment feasibility of the proposed models, an energy efficiency test was conducted under a simulated i.MX6UL edge platform. All baseline and pruned models were executed under identical input and runtime conditions using real sensor data. Power consumption and total energy usage were estimated based on CPU utilization profiles and device thermal design power (TDP) scaling.

The results, summarized in Supplementary Table S2, demonstrate that pruning reduced the overall energy consumption of all six model architectures, with the lowest reduction observed at 12% and the highest at 23%. Beyond energy efficiency, continuous edge operation may also affect hardware longevity due to cumulative thermal stress and component wear. The reduced computational load and power dissipation of the pruned models alleviate these effects by lowering average device temperature and operating strain, thereby supporting sustainable and stable long-term deployment in real storage environments.

In summary, these results showed that the combination of structured pruning and KAN integration provided an effective strategy for compressing deep learning models for spoilage prediction tasks. Among the tested models, the LSTM-KAN model was the most advantageous when prioritizing prediction accuracy, while the TCN-KAN model provided the best balance for resource-limited real-time monitoring systems in postharvest supply chains.

Deployment and Practical Implementation of the AIoT-Enabled Edge Computing System

Based on a comparative evaluation, the TCN-KAN model was identified as the best deployed architecture, achieving the best balance between prediction accuracy and computational efficiency. Its convolutional backbone ensured robust temporal feature extraction, while introducing KAN improved model accuracy and training convergence speed. Before pruning, the TCN-KAN model had the highest precision and optimal training convergence efficiency for deployment in the cloud. A key advantage of choosing TCN-KAN was the adaptability of its architecture to pruning. Compared to LSTM-based architectures, TCN-KAN had relatively small accuracy degradation with parameter pruning and significantly improved the inference speed of the model while maintaining accuracy. This made it particularly suitable for deployment on resource-constrained edge devices for collaborative cloud–edge deployments.

The system was cloud-centric and integrated with distributed edge devices. Unpruned TCN-KAN models were embedded in the cloud, where a Kubernetes (K8s) cluster coordinated computational resources and aggregated data uploaded by multiple edge nodes. The cloud periodically re-trained the model with new data to ensure that the global model adapted to the evolutionary patterns of fruit spoilage and spoilage under different environmental and seasonal conditions. The cloud supported structured pruning of the model to compress the full-scale model into lightweight variants optimized for edge inference. This two-stage process of centralization followed by training and then lightweighting achieved an efficient application between model accuracy and deployment feasibility.

When the model pruning was completed, the updated lightweight models were distributed to the edge devices through the cloud–edge communication interface. These pruned TCN-KAN models performed local, real-time inference with minimal latency, enabling fast spoilage monitoring and localized decision-making without always relying on cloud connectivity. This strategy reduced bandwidth consumption and response time, which was important for monitoring storage environments that required immediate corrective action. Furthermore, this localized processing approach inherently enhances data security and privacy. By performing inference directly on edge devices, the system minimizes the need to transmit high-frequency raw sensor data over external networks, thereby reducing the exposure of sensitive operational data to potential interception. To further ensure system reliability in unattended storage environments, practical fail-safe mechanisms are implemented at the edge. These include real-time data validity checks to filter out transient noise and sensor health monitoring protocols that identify hardware anomalies. In real storage and transportation environments, factors such as fluctuating temperature, humidity variation, and uneven gas diffusion may gradually affect sensor calibration stability and signal consistency. Although the system exhibited stable responses under laboratory conditions, these environmental dynamics could lead to baseline drift or reduced sensitivity over time. Ensuring robust performance under such conditions remains a key challenge for reliable long-term deployment.

The cloud–edge collaboration architecture created a closed-loop system. The cloud served as the central hub for model training, pruning, and version control, while edge devices deployed these optimized models, enabled real-time monitoring and alerting, and uploaded the collected data to the cloud. The endpoint layer consisted of multi-sensor array devices that provided continuous environment monitoring and continuously transmitted data to the edge nodes. This layered design ensured dynamic adaptability: the cloud continuously enhanced the global model based on aggregated data from multiple nodes, while the edge side responded with low latency on a case-by-case basis.

This deployment model worked by combining cloud optimization management with edge response, where the cloud aggregated multi-node data to continuously improve the model, while the edge side provided real-time inference and localized decision-making under resource-constrained conditions. Through this collaborative design, the framework improved spoilage prediction efficiency and informed industrial upgrades in smart warehousing. From a broader socio-economic perspective, the proposed system contributes to more sustainable supply chain management. By providing early and accurate spoilage predictions, it assists stakeholders in making data-driven decisions to significantly reduce postharvest losses and food waste. Additionally, the

adoption of this technology implies a gradual shift in agricultural labor requirements. Rather than replacing human oversight, it facilitates a transition from labor-intensive manual inspections to higher-value roles focused on the management, maintenance, and data analysis of intelligent storage infrastructures, promoting the technical upskilling of the workforce.

Conclusion

An AIoT-based system for predicting apple spoilage in warehousing was developed by integrating multi-sensor data, deep learning architectures, and a scalable cloud–edge–device framework. The key innovation of the system is the introduction of the Kolmogorov–Arnold network (KAN) into traditional temporal models (LSTM, BiLSTM, and TCN), which significantly improves the predictive power and training convergence of the models. Among the evaluated models, the TCN-KAN architecture exhibits the best prediction accuracy and robustness with minimal degradation after pruning, making it particularly suitable for real-time deployment on low-cost, low-computing-power edge devices. By combining these models with a layered cloud–edge architecture, the system achieves efficient, distributed, and adaptive spoilage detection in low-resource environments. Real-time monitoring and alerting are achieved at the edge, while centralized cloud retraining ensures long-term adaptability. The integration of lightweight deployment strategies further improves the responsiveness and practical feasibility of the system. In conclusion, the proposed method provides an accurate, scalable, and cost-effective approach for real-time monitoring and prediction of spoilage in stored apples, highlighting the potential of deep learning and edge intelligence in smart agriculture applications.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11947-026-04228-x>.

Acknowledgements This research was financially supported by the National Key R&D Program of China (Grant No. 2022YFD2100604), the National Natural Science Foundation of China (Grant No. W2412103), and the Zhenjiang Science and Technology Program (Grant No. GJ2024009). The authors gratefully acknowledge these agencies for their generous support.

Author contribution Zhiming Guo: Conceptualization, methodology, supervision, project administration, funding acquisition. Xuebin Ya: software, validation, data curation, writing—original draft preparation, visualization. Chen Yang: formal analysis, investigation. Weixing Sang: formal analysis, investigation. Heera Jayan: writing—review and editing. Chen Wang: resources. Chanjun Sun: resources. Xiaobo Zou: resources, supervision.

Funding This research was funded by the National Key R&D Program of China (Grant No. 2022YFD2100604), the National Natural

Science Foundation of China (W2412103), and the Zhenjiang Science & Technology Program (GJ2024009).

Data Availability Data will be made available on request.

Declarations

Competing Interests The authors declare no competing interests.

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